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E/21/245

CO226 – DATABASE SYSTEMS LAB 04

MOVIE

movie_id	title	year	director
101	Gone with the Wind	1939	Victor Fleming
102	Star Wars	1977	George Lucas
103	The Sound of Music	1965	Robert Wise
104	E.T.	1982	Steven Spielberg
105	Titanic	1997	James Cameron
106	Snow White	1937	NULL
107	Avatar	2009	James Cameron
108	Raiders of the Lost Ark	1981	Steven Spielberg

RATING

reviewer_id	movie_id	stars	rating_date
201	101	2	2011-01-22
202	106	4	NULL
203	103	2	2011-01-20
203	108	4	2011-01-12
204	101	3	2011-01-09
205	103	3	2011-01-27
205	104	2	2011-01-22
205	108	4	NULL
206	106	5	2011-01-19
206	107	3	2011-01-15
207	107	5	2011-01-20
208	104	3	2011-01-02

REVIEWER

reviewer_id	reviewer_name
201	Sarah Martinez
202	Daniel Lewis
203	Brittany Harris
204	Mike Anderson
205	Chris Jackson
206	Elizabeth Thomas
207	James Cameron
208	Ashley White

1. Select **movie ids** of each movie with its **title, reviewer id** and **stars** received.

Query 1

Limit to 1000 rows

```
SELECT M.movie_id, M.title, R.reviewer_id, R.stars
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

movie_id	title	reviewer_id	stars
101	Gone with the Wind	201	2
101	Gone with the Wind	204	3
103	The Sound of Music	203	2
103	The Sound of Music	205	3
104	E.T.	205	2
104	E.T.	208	3
106	Snow White	202	4
106	Snow White	206	5
107	Avatar	206	3
107	Avatar	207	5
108	Raiders of the Lost...	203	4
108	Raiders of the Lost...	205	4

2. Select **movie ids** of each movie with its **title, reviewer id** and **stars** received, where number of stars are less than or equal to three.

Query 1

Limit to 1000 rows

```
SELECT M.movie_id, M.title, R.reviewer_id, R.stars
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
WHERE R.stars <= 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

movie_id	title	reviewer_id	stars
101	Gone with the Wind	201	2
103	The Sound of Music	203	2
101	Gone with the Wind	204	3
103	The Sound of Music	205	3
104	E.T.	205	2
107	Avatar	206	3
104	E.T.	208	3

3. Select each **movie id** with its corresponding **title, reviewer id, reviewer name** and **stars** received.

Query 1

Limit to 1000 rows

```
SELECT M.movie_id, M.title, R.reviewer_id, W.reviewer_name, R.stars
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
JOIN REVIEWER W ON R.reviewer_id = W.reviewer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

movie_id	title	reviewer_id	reviewer_name	stars
101	Gone with the Wind	201	Sarah Martinez	2
101	Gone with the Wind	204	Mike Anderson	3
103	The Sound of Music	203	Brittany Harris	2
103	The Sound of Music	205	Chris Jackson	3
104	E.T.	205	Chris Jackson	2
104	E.T.	208	Ashley White	3
106	Snow White	202	Daniel Lewis	4
106	Snow White	206	Elizabeth Thomas	5
107	Avatar	206	Elizabeth Thomas	3
107	Avatar	207	James Cameron	5
108	Raiders of the Lost...	203	Brittany Harris	4
108	Raiders of the Lost...	205	Chris Jackson	4

4. Select each **movie id** with its corresponding **title, reviewer id, reviewer name** and **stars** received, where the number of **stars** received is equal to five.

Query 1

Limit to 1000 rows

```
SELECT M.movie_id, M.title, R.reviewer_id, W.reviewer_name, R.stars
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
JOIN REVIEWER W ON R.reviewer_id = W.reviewer_id
WHERE R.stars = 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

movie_id	title	reviewer_id	reviewer_name	stars
106	Snow White	206	Elizabeth Thomas	5
107	Avatar	207	James Cameron	5

5. Select movie **title** with its corresponding **reviewer name** and **stars**, where the movie's **rating date** is missing.

Query 1

Limit to 1000 rows

```
SELECT M.title, W.reviewer_name, R.stars
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
JOIN REVIEWER W ON R.reviewer_id = W.reviewer_id
WHERE R.rating_date IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title	reviewer_name	stars
Snow White	Daniel Lewis	4
Raiders of the Lost Ark	Chris Jackson	4

6. Select the details about the reviewers who have a last name called 'Martinez'.

Query 1

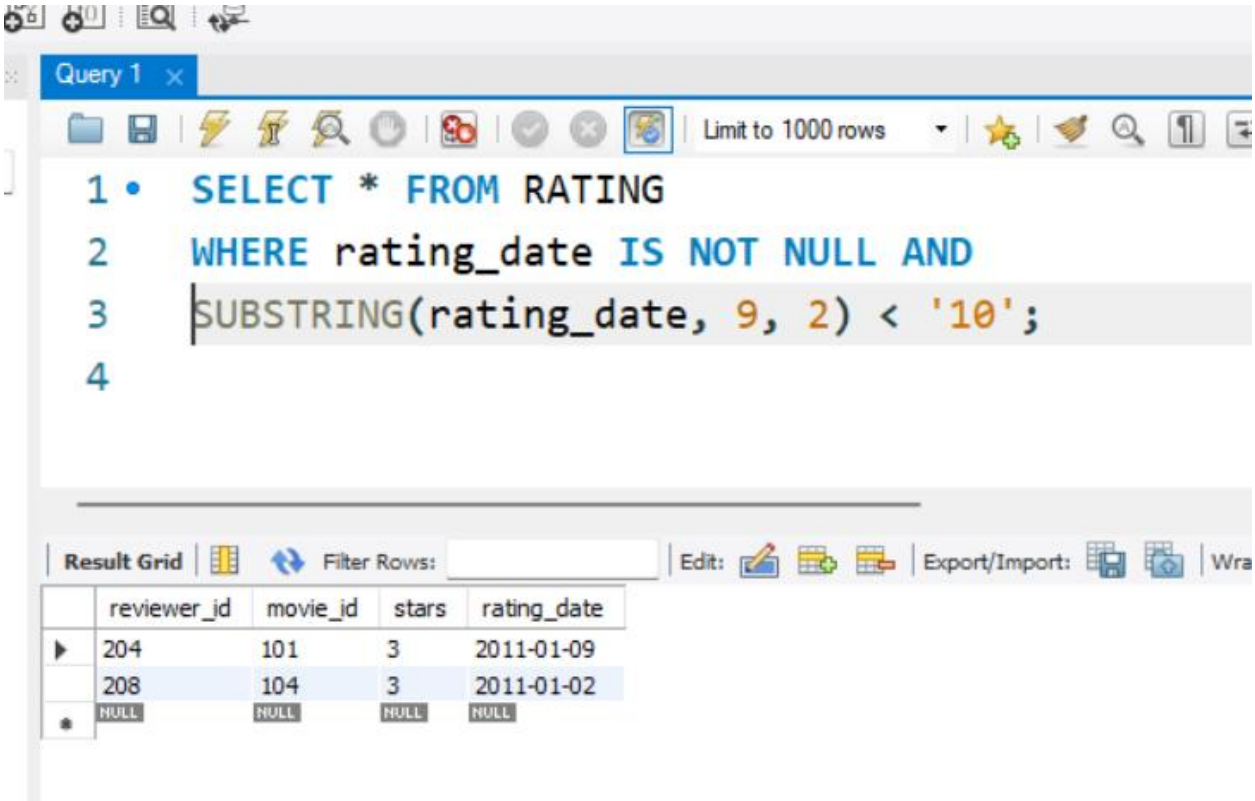
Limit to 1000 rows

```
SELECT * FROM REVIEWER
WHERE reviewer_name LIKE "%Martinez";
```

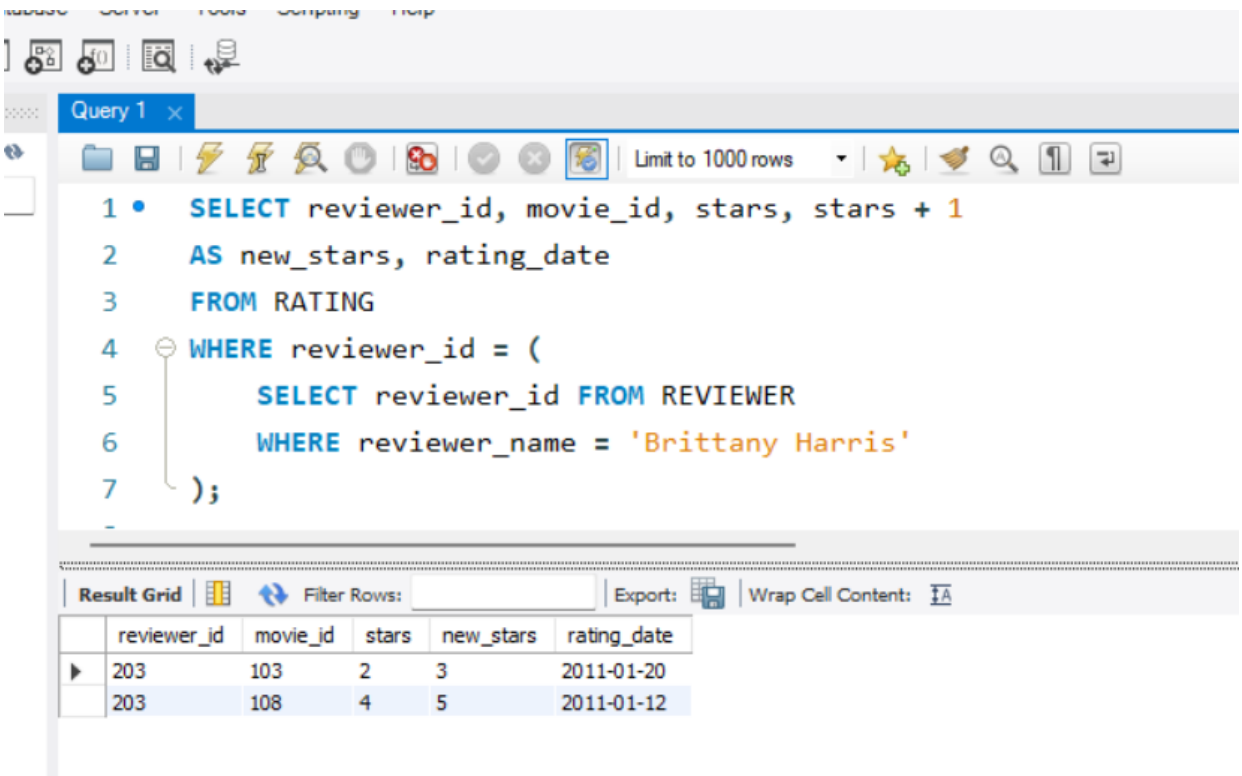
Result Grid | Filter Rows: | Edit: | Export/Import:

reviewer_id	reviewer_name
201	Sarah Martinez
NULL	NULL

7. Select the details about the ratings which have been rated before the 10th day of the month. Use substring comparison.



8. Show the effect of giving one more star to the movies reviewed by 'Brittany Harris'. Here select relevant details from the **RATING** table.



9. Select movie **titles** with its **reviewer name** and **stars** received. Order the result by movie title in the alphabetical order.

```
SELECT M.title, V.reviewer_name, R.stars
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
JOIN REVIEWER V ON R.reviewer_id = V.reviewer_id
ORDER BY M.title;
```

title	reviewer_name	stars
Avatar	Elizabeth Thomas	3
Avatar	James Cameron	5
E.T.	Chris Jackson	2
E.T.	Ashley White	3
Gone with the Wind	Sarah Martinez	2
Gone with the Wind	Mike Anderson	3
Raiders of the Lost Ark	Brittany Harris	4
Raiders of the Lost Ark	Chris Jackson	4
Snow White	Daniel Lewis	4
Snow White	Elizabeth Thomas	5
The Sound of Music	Brittany Harris	2
The Sound of Music	Chris Jackson	3

10. Select movie **titles** with its **stars** received and **rating date**. Order the result by **movie title** in the alphabetical order, then by **stars** and **rating date** both in descending order.

```
SELECT M.title, R.stars, R.rating_date
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
ORDER BY M.title ASC, R.stars DESC, R.rating_date DESC;
```

title	reviewer_name	stars
Avatar	Elizabeth Thomas	3
Avatar	James Cameron	5
E.T.	Chris Jackson	2
E.T.	Ashley White	3
Gone with the Wind	Sarah Martinez	2
Gone with the Wind	Mike Anderson	3
Raiders of the Lost Ark	Brittany Harris	4
Raiders of the Lost Ark	Chris Jackson	4
Snow White	Daniel Lewis	4
Snow White	Elizabeth Thomas	5
The Sound of Music	Brittany Harris	2
The Sound of Music	Chris Jackson	3

11. Write a nested query to list the details of the movies directed by a director
    A. who is also a reviewer.
    B. who is not a reviewer.

A)

```
SELECT * FROM MOVIE
WHERE director IN
    (SELECT reviewer_name FROM REVIEWER);
```

movie_id	title	year	director
105	Titanic	1997	James Cameron
107	Avatar	2009	James Cameron
NULL	NULL	NULL	NULL

B)

```
SELECT * FROM MOVIE
WHERE director NOT IN (SELECT reviewer_name FROM REVIEWER);
```

movie_id	title	year	director
101	Gone with the Wind	1939	Victor Fleming
102	Star Wars	1977	George Lucas
103	The Sound of Music	1965	Robert Wise
104	E.T.	1982	Steven Spielberg
108	Raiders of the Lost Ark	1981	Steven Spielberg
NULL	NULL	NULL	NULL

12. Write a nested query to list the details of the movie ratings,
    A. reviewed by the reviewer 'Sarah Martinez'.
    B. not reviewed by the reviewer 'Sarah Martinez'.

A)

```
SELECT * FROM RATING
WHERE reviewer_id = (
    SELECT reviewer_id FROM REVIEWER
    WHERE reviewer_name = 'Sarah Martinez'
);
```

reviewer_id	movie_id	stars	rating_date
201	101	2	2011-01-22
NULL	NULL	NULL	NULL

B)

Query 1

Limit to 1000 rows

```
SELECT * FROM RATING
WHERE reviewer_id NOT IN (
    SELECT reviewer_id FROM REVIEWER
    WHERE reviewer_name = 'Sarah Martinez'
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

reviewer_id	movie_id	stars	rating_date
202	106	4	NULL
203	103	2	2011-01-20
203	108	4	2011-01-12
204	101	3	2011-01-09
205	103	3	2011-01-27
205	104	2	2011-01-22
205	108	4	NULL
206	106	5	2011-01-19
206	107	3	2011-01-15
207	107	5	2011-01-20
208	104	3	2011-01-02
NULL	NULL	NULL	NULL

13. Find all the years that have a movie that received a rating of 4 or 5 and sort them in increasing order of the year. Write,
    A. a non-nested query.
    B. a nested query.

A)

Query 1

Limit to 1000 rows

```
SELECT DISTINCT M.year
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
WHERE R.stars >= 4
ORDER BY M.year;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year
1937
1981
2009

B)

Query 1

Limit to 1000 rows

```
SELECT DISTINCT year
FROM MOVIE
WHERE movie_id IN (
    SELECT movie_id FROM RATING
    WHERE stars >= 4
)
ORDER BY year;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

year
1937
1981
2009

14. Some reviewers did not provide a date with their rating. Find the names of all reviewers who have a NULL value for the date. Write a nested query.

Query 1

Limit to 1000 rows

```
SELECT reviewer_name
FROM REVIEWER
WHERE reviewer_id IN (
    SELECT reviewer_id FROM RATING
    WHERE rating_date IS NULL
);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

reviewer_name
Daniel Lewis
Chris Jackson

15. List the **movie titles** and average **ratings**, from the highest-rated to lowest-rated. If two or more movies have the same average rating, list them in alphabetical order.

Query 1

Limit to 1000 rows

```
SELECT M.title, AVG(R.stars) AS avg_rating
FROM MOVIE M
JOIN RATING R ON M.movie_id = R.movie_id
GROUP BY M.title
ORDER BY avg_rating DESC, M.title ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

title	avg_rating
Snow White	4.5000
Avatar	4.0000
Raiders of the Lost Ark	4.0000
E.T.	2.5000
Gone with the Wind	2.5000
The Sound of Music	2.5000

16. Remove all ratings where the movie's year is before 1970 or after 2000

Query 1

Limit to 1000 rows

```
DELETE FROM RATING
WHERE movie_id IN (
    SELECT movie_id FROM MOVIE
    WHERE year < 1970 OR year > 2000
);
```

17. Remove all ratings where the rating date is NULL

Query 1

Limit to 1000 rows

```
DELETE FROM RATING
WHERE rating_date IS NULL;
```